M.Sc. in IT – Enterprise Application Development
2026 - Year 01 – Semester 01
SE5070 – Enterprise Mobility

Assignment

Objective : The objective of this project is to demonstrate the knowledge and the expertise of mobile application development.

Percentage of overall : 30%
Duration : 08 Week *(Take home assignment)*
Assignment Type : Individual
Deadline : 18th October 2026
Assignment created by : Lasitha Petthawadu

Project Submission

Project document needs to be submitted to Moodle as a softcopy with a link to the source code on Github. Students are expected to participate in a live viva. Participation for the viva is mandatory and not participating will result in zero marks.

Assignment Outline

Students are required to formulate a mobile project idea which will fall under one of the following areas.

0. Edtech
1. Entertainment
2. Sustainability & GreenTech
3. Artificial Intelligence
4. HealthTech

To figure out your project topic a student needs to take their index number and add all the numbers of the index number (calculate the digit sum) and find the modulus by 5, the resulting number would denote your respective project area.

Once the project area is identified, students are required to come up with a mobile project idea in this area. When formulating the project idea keep in mind on the time duration and focus on identifying features that represent the following

- Students can use any compile to native framework for development (React Native, Flutter) or even native android or IOS
- The project should interact with a backend service. Some of the choices for the backend could be, Google Firebase (https://firebase.google.com/), AWS Amplify or even a custom backend to serve the data needed by the application.
- The application should support offline capabilities. And the application should sync back the offline data once the application comes back online or receives network coverage.
- The application needs to focus on constraint design algorithms and principles.

- Integration of native hardware of choice (Camera, GPS, Gyro)
- Should have one or more custom control/component(s) implemented by the student himself (This doesn't mean that the entire application components need to be custom components! Students can use a component library of choice. But there should be custom components developed by the student as well, as part of the application).

Assignment Submissions

A project document needs to be prepared outlining the following;

- The calculation of the digit sum of the index number and the project area based off of it.
- An overview of the project idea and the features.
- The architecture and design of the application.
    - Outline major decisions taken when designing the application.
    - Outline how data flow and interaction works between layers (Make use of the knowledge from lectures.)
    - Outline how the source is organized and structured in the application and explain why it is done so.
- How offline access works in the application and techniques used.
- Outline of the constraint design algorithms and principles.
- Outline of how native hardware integrates to the
- Outline of the UI/UX flow with screenshots of the application.
- Outline of the custom components and how it functions. Explain the properties and events defined in the component(s).
- Outline of Decision Records & Failure Logs - You need to document your process of implementation if you used AI for certain parts you need to provide prompts and show the failures and errors that you got during your implementation.

Presentation and Viva

- Live Vivas will be conducted maximum 20 minutes
- Live Coding Defense - You will be asked to code / implement a simple part in your application
- Be prepared with your demonstration
- Viva schedule will be created and no last minute rescheduling

Evaluation Criteria

The project would be evaluated based on the following criteria. Make sure to distribute the effort wisely based on the weightage of the criteria outlined below.

Criteria	Weightage
Uniqueness and creativeness of the project Idea in the assigned domain area. • Able to rationalize why the idea is worth-while. • Documentation submitted	15%
Solidness of the implementation ● Source code structuring and in-depth understanding of how data flow works in the application. ● Application Design, decisions made. ● Offline Data access implementation. ● Constrained app design principles. ● Completeness of the application. ● Live coding defense	50%
Richness of the User experience and the Navigation ● Rational behind UI/UX selections and flows. ● Components used and Custom UI components implemented.	20%
Presentation and the solution (Soft-skills) ● Confidence in the work done ● Communication skills ● Intuitive thinking	15%

Code of Conduct

Students are expected to adhere to the following guidelines.

Constraints:

1. Should any plagiarism be identified during viva, the assignment will be zeroed
2. Students who get any category need to come up with assets of the application such as images, game assets on their own and should not copy any existing codebase, or should not take an existing collision logic or any codebase that is readily available. If any of such is identified then the assignment will be marked as zero.
3. Students are encouraged to help others and to support batch mates. However, if done so, please mention who you helped and what type of help was provided, along with their index number of the student so that the student would be excluded from plagiarism.